

QA Dataset Cleaning & Standardization Log (Two-Stage Pipeline)

This document provides a detailed, step-by-step audit log of all data cleaning, structural repairs, formatting corrections, and label standardizations applied across the two stages of our dataset cleaning pipeline.

Stage 1: Raw CSV → Standardized Dataset (`updated3.0.csv`)

This stage covers all casing consolidations, spelling corrections, date/time format normalizations, structural cell repairs, and basic status filtering.

1. Row Volume Changes

- **Initial Raw Rows:** 5,110 rows.
- **Rows Filtered Out:** 2,162 rows.
- ***Rejection Logic:** Removed rows where `Overall Test Status` was empty/NA or `Type of Question` was empty/NA.
- **Resulting Rows:** 2,948 rows.

2. Date & Time Standardizations

- **Date Format Unification:** Converted mixed date strings (e.g. `10.06.2026`, `14-06-2026`, or spacing errors like `12-06 -2026`) to standard ISO `YYYY-MM-DD`.
- **Time Normalization (Asked & Received Columns):**
 - Unified float decimal representation (e.g., Asked time `11.0` → `11:00:00`, Received time `6.52` → `06:52:00`).
 - Removed text string suffixes and commas (e.g., `06:52:AM` → `06:52:00`, `, 05:01 PM` → `17:01:00`).
 - Converted 12-hour AM/PM times (e.g., `2:32:32 PM`) to 24-hour equivalent (`14:32:32`).

3. Actor Name Standardizations

Consolidated tester, author, reviewer, and moderator names to Title Case, stripping trailing periods, newlines, and extra spaces. Mapped raw variations to standard distinct names:

- **CH. Sharmila:** Mapped from `CH.Sharmila`, `CH.sharmila`, `CH. sharmila`
- **Dhaarani S:** Mapped from `Dhaarani S`, `Dhaarani. S`, `Dhaarani.S`, `Dhaarani.S`, `Dhaarani. S`
- **Ithagani Shireesha:** Mapped from `Ithagani Shireesha`, `Ithagani shireesha`, `Ithagani shireesha`, `Ithagani Shireesha`
- **Joydeep:** Mapped from `JOYDEEP`, `JOYDEEP`, `JHOYDEEP`

- **K. Deni Sudha:** Mapped from `K.Deni sudha`, `K. Deni sudha`, `K.Deni sudha`
- **Lavanya Mathialagan:** Mapped from `Lavanya Mathialagan`, `Lavanya \nmathialagan`, `Lavanya \nMathialagan`, `lavanya Mathialagan`, `Lavanya Mathialagan`, `Lavanya Mathialagan`, `Lavanya mathialagan`
- **Rishi Kumar G:** Mapped from `Rishi Kumar G`, `Rishi Kumar.G`
- **T. Vishnu Vardhan:** Mapped from `T.Vishnu Vardhan`, `T.Vishnu vardhan`, `T. Vishnu vardhan`, `T. Vishnu Vardhn`, `T. Vishnu Vardhan`, `T.Vishnu Vardhan`
- **B. Sasidhar:** Mapped from `B.Sasidhar`, `B.sasidhar`, `SASHIDHAR`, `SASIDHAR`, `Sasidhar`, `b.sasidhar`, `sasidhar`
- **Jahnavi Gaddam:** Mapped from `JAHANVI`, `JAHNAVI`, `JAHNAVI.G`, `JAHNAVI.GADDAM`, `JAHNVAI`, `JANHNI`, `JANHVI`, `JANHVI`, `Jahanavi`, `Jahnavi`, `Janahavi`, `Janhavi`, `Jhanavi`, `Jhanvai`, `jahnavi`
- **Mohammad Saleem Paasha:** Mapped from `MOHAMMAD`, `MOHAMMAND`, `MOHAMMED SALEEM PAASHA`, `Mohammad`, `Mohammad Saleem pasha`, `Mohammed`, `Mohammad`, `md.saleem`, `saleem`

4. Structural Row & Cell Repair Log

- **Row `TL-2477`:** Swapped `Reviewer1 Name` and `Reviewer1 Assignment Time` to put name `Ambika` and timestamp `11:39:47 Am` into their respective columns, fixing the transposed cells.
- **Row `TL-5222`:** Realigned the shifted cell block in columns 39-43 to restore Reviewer4's name (`Suraiya Amin`) and datetimes to their correct column offsets.

5. Binary & Multi-Class Labels

- **Overall Test Status:** Mapped `PASS`, `pass` → `Pass`.
- **Defect Severity:** Mapped `NO Defect`, `no dfect` → `No Defect`. Standardized spelling typos (`critical` → `Critical`).
- **Yes/No Columns (Casing Mappings):** Standardized casing variants and spaces (e.g. `yes`/`YES`/`y` → `Yes`; `no`/`NO`/`n` → `No`) across all binary columns.
- **Category Names:** Consolidated spelling and wording overlaps (e.g. `Fertiliser use & availability` → `Fertilizer Use and Availability`).

Stage 2: Standardized Dataset (`updated3.0.csv`) → Final Cleaned Dataset (`cleandataset.csv`)

This stage covers the strict parameter filtering where rows were removed if any core KPI calculation data was missing.

1. Row Volume Changes

- **Standardized Rows** (`updated3.0.csv`): 2,948 rows.
- **Rows Filtered Out:** 2,039 rows.
- ***Rejection Logic:** Excluded any row where any core dashboard parameter was missing (empty/NIL/NA).
- **Final Clean Rows:** 909 rows.

2. Core Parameters Enforced (Must NOT be Null)

Rows were kept only if they had valid, non-empty data in the following **18 core columns**:

- `Test Date`, `Tester Name`, `Build/Version`, `Channel Tested`, `Language Tested`, `Question ID`, `Question Category`, `Type of Question`, `Overall Test Status`, `SLA Status`, `Defect Severity`, `Answer Scientifically Correct?`, `Question Saved in DB?`, `Answer Saved in DB?`, `Q-ID Consistent Across Systems?`, `Source Links Provided?`, `Translation Quality`, and `Response Time (mins) [Auto]`.

3. Conditional Rules Enforced (Null-Filtering)

- **Category-Specific Correctness:**
 - If Category was `Weather Queries`, `Weather Q Answered Correctly?` was verified and required.
 - If Category was `Mandi Price Queries` or `Market Prices, MSP and Marketing`, `Mandi Price Q Correct?` was verified and required.
 - If Category was `Scheme Queries`, `Scheme Q Correct?` was verified and required.
- **Expert Display vs Expert Name Matching:**
 - If `Expert Name Displayed?` was `"Displayed"`, we verified and required that `Correct Expert Name?` was not empty. Any row where expert displayed was `"Displayed"` but `Correct Expert Name?` was null was excluded.

Executive KPI Column Mapping Specification

Executive KPI	Mapped Spreadsheet Columns
ACE TRUST SCORE	Answer Scientifically Correct?, Correct Expert Name?, Expert Name Displayed?, Source Links Provided?, Translation Quality, Weather Q Answered Correctly?, Mandi Price Q Correct?, Scheme Q Correct?, WhatsApp vs Web Answer Match?, Question Correctly Framed?
FARMER EXPERIENCE SCORE	Response Time (mins) [Auto], SLA Status, Question Correctly Framed?, Translation Quality, Voice Input Working?, Voice Output Working?, Voice Input Quality, Voice Output Quality, Notification Received?, Notification on Same Thread?, Notification Linked Correct Q-ID?, 120-min Msg Shown to User?
CRITICAL FAILURES TODAY	Answer Scientifically Correct?, Weather Q Answered Correctly?, Mandi Price Q Correct?, Scheme Q Correct?, Question Saved in DB?, Answer Saved in DB?, Q-ID Consistent Across Systems?, Notification Linked Correct Q-ID?, Defect Severity
RELEASE HEALTH	Overall Test Status, Defect Severity, Status, Build / Version, Sprint / Cycle, Question Saved in DB?, Answer Saved in DB?, Q-ID Consistent Across Systems?
BIGGEST BOTTLENECK	Author TAT, Review1 TAT, Review2 TAT, Review3 TAT, Review4 TAT, Review5 TAT, Moderator TAT
WEAKEST MODULE	Weather Q Answered Correctly?, Mandi Price Q Correct?, Scheme Q Correct?, Translation Quality, Voice Input Quality, Voice Output Quality
TREND	Trust Score trend, Farmer Experience trend, Response Time trend, Defect trend, Review TAT trend